(a) On Diagram 1, sketch a graph of the curve $C$ with equation

$$y = \frac{6}{x} \qquad x \neq 0$$

**(2)**

The curve $C$ is transformed onto the curve with equation $y = \frac{6}{x-2} \qquad x \neq 2$

(b) Fully describe this transformation.

**(2)**

The curve with equation

$$y = \frac{6}{x-2} \qquad x \neq 2$$

and the line with equation

$$y = kx + 7 \qquad \text{where } k \text{ is a constant}$$

intersect at exactly two points, $P$ and $Q$.

Given that the $x$ coordinate of point $P$ is $-4$

(c) find the value of $k$,

**(2)**

(d) find, using algebra, the coordinates of point $Q$.

(*Solutions relying entirely on calculator technology are not acceptable.*)

**(4)**